Equality-Constrained Optimization

Lagrange's Theorem and the Second-Order Necessary Conditions

Theorems 20.3 and 20.4

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The goal

The two central necessary conditions for the problem

$$\text{minimize } f(\boldsymbol{x}) \qquad \text{subject to } \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}.$$

- **Theorem 20.3 (Lagrange's Theorem)** — a *first-order* necessary condition: $Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}^\top$.
- **Theorem 20.4 (Second-Order Necessary Conditions)** — adds curvature information: $\boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{y} \geq 0$ for all $\boldsymbol{y}$ in the tangent space.

Guiding idea

A constrained problem is turned into an *unconstrained* problem in one variable by *walking along a curve inside the constraint set*. All the familiar single-variable calculus ($\varphi'(t^*) = 0$, $\varphi''(t^*) \geq 0$) is then applied to that walk.

Outline

The problem

Equality-constrained problem

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}, \end{aligned}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $f : \mathbb{R}^n \to \mathbb{R}$, and $\boldsymbol{h} = [h_1, \ldots, h_m]^\top : \mathbb{R}^n \to \mathbb{R}^m$ with $m \leq n$.

- Writing the constraint componentwise: $h_1(\boldsymbol{x}) = 0, \ \ldots, \ h_m(\boldsymbol{x}) = 0$.
- Standing smoothness assumption: $\boldsymbol{h} \in \mathcal{C}^1$ (for Theorem 20.3); $f, \boldsymbol{h} \in \mathcal{C}^2$ (for Theorem 20.4).
- **Why $m \leq n$?** With more independent equations than unknowns the feasible set is generically empty; with $m \leq n$ we expect an $(n - m)$-dimensional feasible set.
- Maximization needs no separate theory: $\max f = -\min(-f)$, and the necessary conditions of Theorem 20.3 hold for maximizers as well.

Notation: gradients, Jacobians, Hessians

For $f : \mathbb{R}^n \to \mathbb{R}$,

$$Df(\boldsymbol{x}) = \left[\tfrac{\partial f}{\partial x_1}(\boldsymbol{x}), \ldots, \tfrac{\partial f}{\partial x_n}(\boldsymbol{x})\right] \ \ (1 \times n \textit{ row } \text{vector}), \qquad \nabla f(\boldsymbol{x}) = Df(\boldsymbol{x})^\top \ \ (n \times 1).$$

For $\boldsymbol{h} : \mathbb{R}^n \to \mathbb{R}^m$ the **Jacobian** is the $m \times n$ matrix

$$D\boldsymbol{h}(\boldsymbol{x}) = \begin{bmatrix} Dh_1(\boldsymbol{x}) \\ \vdots \\ Dh_m(\boldsymbol{x}) \end{bmatrix} = \begin{bmatrix} \nabla h_1(\boldsymbol{x})^\top \\ \vdots \\ \nabla h_m(\boldsymbol{x})^\top \end{bmatrix}.$$

Hessians ($n \times n$, symmetric when the function is $\mathcal{C}^2$):

$$\boldsymbol{F}(\boldsymbol{x}) = D^2 f(\boldsymbol{x}) = \left[\tfrac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{x})\right], \qquad \boldsymbol{H}_k(\boldsymbol{x}) = D^2 h_k(\boldsymbol{x}), \quad k = 1, \ldots, m.$$

A recurring habit

Row vector $\times$ column vector: $Df(\boldsymbol{x})\boldsymbol{y} = \nabla f(\boldsymbol{x})^\top \boldsymbol{y} = \langle \nabla f(\boldsymbol{x}), \boldsymbol{y} \rangle$. Keep track of shapes — every step below is a shape-consistent matrix identity.

The feasible set is a surface

Definition (surface defined by the constraints)

$$S = \{\boldsymbol{x} \in \mathbb{R}^n \ : \ h_1(\boldsymbol{x}) = 0, \ldots, h_m(\boldsymbol{x}) = 0\} = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}\}.$$

Definition 20.2 (regular point)

A point $\boldsymbol{x}^*$ satisfying $\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}$ is a **regular point** of the constraints if the gradient vectors

$$\nabla h_1(\boldsymbol{x}^*), \ \nabla h_2(\boldsymbol{x}^*), \ \ldots, \ \nabla h_m(\boldsymbol{x}^*)$$

are *linearly independent*. Equivalently, rank $D\boldsymbol{h}(\boldsymbol{x}^*) = m$ (full row rank).

- If every point of S is regular, S is a smooth surface of dimension $n - m$.
- **Regularity is a condition on the description of S, not on S itself** — the same set can be described regularly or non-regularly.

Curves on a surface

Definition 20.3 (curve)

A **curve** C on a surface S is a set of points $\{\boldsymbol{x}(t) \in S : \ t \in (a, b)\}$, continuously parameterised by t; i.e. $\boldsymbol{x} : (a, b) \to S$ is continuous. The curve *passes through* $\boldsymbol{x}^*$ if $\boldsymbol{x}(t^*) = \boldsymbol{x}^*$ for some $t^* \in (a, b)$.

Definition 20.4 (differentiable / twice differentiable curve)

C is **differentiable** if $\dot{\boldsymbol{x}}(t) = \frac{d\boldsymbol{x}}{dt}(t) = [\dot{x}_1(t), \ldots, \dot{x}_n(t)]^\top$ exists for all $t \in (a, b)$; it is **twice differentiable** if $\ddot{\boldsymbol{x}}(t)$ exists for all $t \in (a, b)$.

Definition 20.5 (tangent space)

The **tangent space** at a point $\boldsymbol{x}^*$ on the surface $S = \{\boldsymbol{x} : \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}\}$ is

$$T(\boldsymbol{x}^*) = \{\boldsymbol{y} \in \mathbb{R}^n \ : \ D\boldsymbol{h}(\boldsymbol{x}^*)\boldsymbol{y} = \boldsymbol{0}\} = \mathcal{N}\big(D\boldsymbol{h}(\boldsymbol{x}^*)\big).$$

Tangent space: a computed example

$S = \{\mathbf{x} \in \mathbb{R}^3 : h_1(\mathbf{x}) = x_1 = 0,\ \ h_2(\mathbf{x}) = x_1 - x_2 = 0\}$

$$D\mathbf{h}(\mathbf{x}) = \begin{bmatrix} \nabla h_1(\mathbf{x})^\top \\ \nabla h_2(\mathbf{x})^\top \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix}, \qquad \text{rank} = 2 \ \Rightarrow \ \text{all points regular.}$$

$$T(\mathbf{x}) = \left\{ \mathbf{y} : \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \mathbf{0} \right\} = \{[0, 0, \alpha]^\top : \alpha \in \mathbb{R}\} = \text{the } x_3\text{-axis.}$$

Here S *is* the x_3-axis, and $\dim T = n - m = 3 - 2 = 1$. ✓

Note: the tangent space is computed by pure linear algebra — solve a homogeneous linear system whose coefficient matrix is the Jacobian of the constraints.

Theorem 20.1: the tangent space really consists of tangent vectors

Theorem 20.1

Suppose $\boldsymbol{x}^* \in S$ is a *regular* point and $T(\boldsymbol{x}^*)$ is the tangent space at $\boldsymbol{x}^*$. Then $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ *if and only if* there is a differentiable curve in S passing through $\boldsymbol{x}^*$ with derivative $\boldsymbol{y}$ at $\boldsymbol{x}^*$.

Proof of "$\Leftarrow$" (the easy direction — and the one we reuse)

Let $\{\boldsymbol{x}(t)\} \subset S$ with $\boldsymbol{x}(t^*) = \boldsymbol{x}^*$, $\dot{\boldsymbol{x}}(t^*) = \boldsymbol{y}$. Since the curve lies in S,

$$\boldsymbol{h}(\boldsymbol{x}(t)) = \boldsymbol{0} \quad \text{for all } t \xrightarrow{\frac{d}{dt},\ \text{chain rule}} D\boldsymbol{h}(\boldsymbol{x}(t))\,\dot{\boldsymbol{x}}(t) = \boldsymbol{0}.$$

At $t = t^*$: $D\boldsymbol{h}(\boldsymbol{x}^*)\boldsymbol{y} = \boldsymbol{0}$, i.e. $\boldsymbol{y} \in T(\boldsymbol{x}^*)$. ∎

Where regularity enters

The converse "$\Rightarrow$" (*every* $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ is realised by an actual curve on S) is the deep half: it is proved with the **Implicit Function Theorem** and *requires* $\operatorname{rank} D\boldsymbol{h}(\boldsymbol{x}^*) = m$ — **the only place regularity is used in Theorem 20.3.**

Normal space

Definition 20.6 (normal space)

The **normal space** at $\boldsymbol{x}^* \in S$ is

$$N(\boldsymbol{x}^*) = \{\boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{x} = D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{z}, \ \boldsymbol{z} \in \mathbb{R}^m\} = \mathcal{R}\big(D\boldsymbol{h}(\boldsymbol{x}^*)^\top\big).$$

Expanding the matrix–vector product $D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{z}$ column by column:

$$N(\boldsymbol{x}^*) = \operatorname{span}\left[\nabla h_1(\boldsymbol{x}^*), \ldots, \nabla h_m(\boldsymbol{x}^*)\right] = \left\{\boldsymbol{x} = \textstyle\sum_{i=1}^m z_i \nabla h_i(\boldsymbol{x}^*) \ : \ z_i \in \mathbb{R}\right\}.$$

- $N(\boldsymbol{x}^*)$ is a subspace containing $\mathbf{0}$; if $\boldsymbol{x}^*$ is regular, $\dim N(\boldsymbol{x}^*) = m$.
- **Read it as**: the normal space is exactly the set of all linear combinations of the constraint gradients. *This is the set in which Lagrange's theorem will place $\nabla f(\boldsymbol{x}^*)$.*
- Normal plane: $NP(\boldsymbol{x}^*) = N(\boldsymbol{x}^*) + \boldsymbol{x}^*$.

The bridge between T and N

$$T(\boldsymbol{x}^*) = N(\boldsymbol{x}^*)^\perp \qquad \text{and} \qquad T(\boldsymbol{x}^*)^\perp = N(\boldsymbol{x}^*).$$

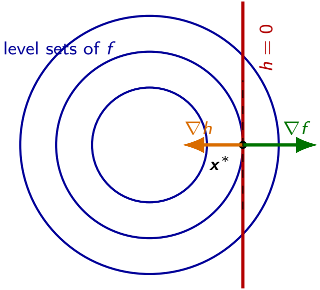
Consequence used constantly

$\mathbb{R}^n = N(\boldsymbol{x}^*) \oplus T(\boldsymbol{x}^*)$: every $\boldsymbol{v} \in \mathbb{R}^n$ splits *uniquely* as $\boldsymbol{v} = \boldsymbol{w} + \boldsymbol{y}$ with $\boldsymbol{w} \in N(\boldsymbol{x}^*)$, $\boldsymbol{y} \in T(\boldsymbol{x}^*)$. In particular

$$\boxed{\nabla f(\boldsymbol{x}^*) \perp T(\boldsymbol{x}^*) \iff \nabla f(\boldsymbol{x}^*) \in N(\boldsymbol{x}^*)}$$

which is the whole content of Lagrange's theorem in geometric form.

Why should ∇f and ∇h be parallel? $(n = 2,\ m = 1)$



Let $\boldsymbol{x}^*$ minimise f on the curve $\{h = 0\}$ and parameterise the curve by $\boldsymbol{x}(t)$ with $\boldsymbol{x}(t^*) = \boldsymbol{x}^*$, $\dot{\boldsymbol{x}}(t^*) \neq \mathbf{0}$.

1. $h(\boldsymbol{x}(t)) = 0 \ \Rightarrow\ \nabla h(\boldsymbol{x}^*)^\top \dot{\boldsymbol{x}}(t^*) = 0$:
$$\nabla h(\boldsymbol{x}^*) \perp \dot{\boldsymbol{x}}(t^*).$$
2. $\varphi(t) = f(\boldsymbol{x}(t))$ has a minimum at t^*, so $\varphi'(t^*) = \nabla f(\boldsymbol{x}^*)^\top \dot{\boldsymbol{x}}(t^*) = 0$:
$$\nabla f(\boldsymbol{x}^*) \perp \dot{\boldsymbol{x}}(t^*).$$
3. In $\mathbb{R}^2$, the orthogonal complement of a single nonzero vector is a *line*; two vectors orthogonal to the same nonzero vector must be **parallel**.

Theorem 20.2 (Lagrange's theorem, $n = 2$, $m = 1$)

If $\boldsymbol{x}^*$ minimises $f : \mathbb{R}^2 \to \mathbb{R}$ subject to $h(\boldsymbol{x}) = 0$ and $\nabla h(\boldsymbol{x}^*) \neq \mathbf{0}$, then there is a scalar λ^* with $\nabla f(\boldsymbol{x}^*) + \lambda^* \nabla h(\boldsymbol{x}^*) = \mathbf{0}$.

From $\mathbb{R}^2$ to $\mathbb{R}^n$: what has to change

- In $\mathbb{R}^2$ with one constraint we used "orthogonal to the same vector $\Rightarrow$ parallel". **That argument is unavailable when $n > 2$ or $m > 1$**: the orthogonal complement of $\dot{\boldsymbol{x}}(t^*)$ is then $(n-1)$-dimensional and contains many non-parallel vectors.
- The correct general replacement is the pair of subspaces $T(\boldsymbol{x}^*), N(\boldsymbol{x}^*)$:

$$\underbrace{\nabla f(\boldsymbol{x}^*) \perp \textit{every}\ \text{tangent direction}}_{\text{calculus, one curve at a time}} \iff \underbrace{\nabla f(\boldsymbol{x}^*) \in N(\boldsymbol{x}^*) = \operatorname{span}[\nabla h_i(\boldsymbol{x}^*)]}_{\text{linear algebra, Lemma 20.1}}$$

- "Parallel to ∇h" becomes "**a linear combination of the ∇h_i**";
- the single scalar λ^* becomes a **vector $\boldsymbol{\lambda}^* \in \mathbb{R}^m$** of multipliers.
- To know that *every* $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ is the velocity of an actual curve on S we need Theorem 20.1 — and therefore **regularity**.

Theorem 20.3 (Lagrange's Theorem)

Statement

Let $\boldsymbol{x}^*$ be a **local minimizer (or maximizer)** of $f : \mathbb{R}^n \to \mathbb{R}$, subject to $\boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}$, where $\boldsymbol{h} : \mathbb{R}^n \to \mathbb{R}^m$, $m \leq n$. Assume that $\boldsymbol{x}^*$ is a **regular point** of the constraints. Then there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ such that

$$\boxed{Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}^\top}$$

Equivalent formulations (transpose the row-vector identity):

$$\nabla f(\boldsymbol{x}^*) + D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}^* = \boldsymbol{0} \qquad \Longleftrightarrow \qquad \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{m} \lambda_i^* \nabla h_i(\boldsymbol{x}^*) = \boldsymbol{0}.$$

- $\boldsymbol{\lambda}^*$ is the **Lagrange multiplier vector**; its entries are the *Lagrange multipliers*.
- Together with the feasibility equation $\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}$ this pair is called the **Lagrange condition**: $n + m$ equations in the $n + m$ unknowns $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$.

Proof of Theorem 20.3 — the strategy first

We must produce a vector $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ with $\nabla f(\boldsymbol{x}^*) = -D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}^*$. Observe what that statement *means*:

$$\exists\, \boldsymbol{\lambda}^* : \ \nabla f(\boldsymbol{x}^*) = D\boldsymbol{h}(\boldsymbol{x}^*)^\top(-\boldsymbol{\lambda}^*) \iff \nabla f(\boldsymbol{x}^*) \in \mathcal{R}\big(D\boldsymbol{h}(\boldsymbol{x}^*)^\top\big) = N(\boldsymbol{x}^*).$$

So the theorem is equivalent to a membership statement

Goal: $\quad \nabla f(\boldsymbol{x}^*) \in N(\boldsymbol{x}^*).$

By **Lemma 20.1**, $N(\boldsymbol{x}^*) = T(\boldsymbol{x}^*)^\perp$. Hence it suffices to prove

Reduced goal: $\quad \nabla f(\boldsymbol{x}^*) \in T(\boldsymbol{x}^*)^\perp, \quad$ i.e. $\quad \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{y} = 0$ for every $\boldsymbol{y} \in T(\boldsymbol{x}^*)$.

The logical shape of the proof

"*Existence of multipliers*" (hard to construct directly) has been converted into "*orthogonality against a subspace*" (checkable one vector at a time, by calculus). This is the single most important idea in the proof.

Proof of Theorem 20.3 — Steps 1 and 2

Step 1. Fix an arbitrary tangent direction

Let $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ be arbitrary. Since $\boldsymbol{x}^*$ is a **regular** point, Theorem 20.1 (direction "$\Rightarrow$") gives a *differentiable curve* $\{\boldsymbol{x}(t) : t \in (a, b)\}$ lying in S with

$$\boldsymbol{h}(\boldsymbol{x}(t)) = \boldsymbol{0} \;\; \forall t \in (a, b), \qquad \boldsymbol{x}(t^*) = \boldsymbol{x}^*, \qquad \dot{\boldsymbol{x}}(t^*) = \boldsymbol{y}$$

for some $t^* \in (a, b)$.

Step 2. Restrict f to the curve: an unconstrained 1-D problem

Define the composite function $\varphi : (a, b) \to \mathbb{R}$, $\varphi(t) = f(\boldsymbol{x}(t))$.

Claim: t^* is an unconstrained local minimizer of φ. Indeed, $\boldsymbol{x}^*$ minimises f over $S \cap B(\boldsymbol{x}^*, \varepsilon)$ for some $\varepsilon > 0$; by continuity of $\boldsymbol{x}(\cdot)$ there is $\delta > 0$ with $\|\boldsymbol{x}(t) - \boldsymbol{x}^*\| < \varepsilon$ whenever $|t - t^*| < \delta$. Since $\boldsymbol{x}(t) \in S$ always,

$$\varphi(t) = f(\boldsymbol{x}(t)) \;\geq\; f(\boldsymbol{x}^*) = \varphi(t^*) \qquad \text{for all } |t - t^*| < \delta.$$

Proof of Theorem 20.3 — Steps 3 and 4

Step 3. Apply the unconstrained first-order necessary condition

φ is differentiable and has an interior local minimum at t^*, so (Theorem 6.1, the one-variable FONC)

$$\frac{d\varphi}{dt}(t^*) = 0.$$

Step 4. Compute that derivative by the chain rule

$$\frac{d\varphi}{dt}(t) = \frac{d}{dt} f(\boldsymbol{x}(t)) = Df(\boldsymbol{x}(t))\, \dot{\boldsymbol{x}}(t).$$

Evaluating at $t = t^*$, where $\boldsymbol{x}(t^*) = \boldsymbol{x}^*$ and $\dot{\boldsymbol{x}}(t^*) = \boldsymbol{y}$:

$$0 = \frac{d\varphi}{dt}(t^*) = Df(\boldsymbol{x}^*)\, \boldsymbol{y} = \nabla f(\boldsymbol{x}^*)^\top \boldsymbol{y}.$$

Since $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ was arbitrary: $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{y} = 0$ for all $\boldsymbol{y} \in T(\boldsymbol{x}^*)$.

Step 5. Translate orthogonality back into multipliers

Step 4 says exactly

$$\nabla f(\boldsymbol{x}^*) \in T(\boldsymbol{x}^*)^\perp \overset{\text{Lemma 20.1}}{=} N(\boldsymbol{x}^*) \overset{\text{Def. 20.6}}{=} \mathcal{R}(D\boldsymbol{h}(\boldsymbol{x}^*)^\top).$$

By definition of the range, there exists a vector $\boldsymbol{z} \in \mathbb{R}^m$ with $\nabla f(\boldsymbol{x}^*) = D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{z}$. Put $\boldsymbol{\lambda}^* := -\boldsymbol{z}$. Then

$$\nabla f(\boldsymbol{x}^*) + D\boldsymbol{h}(\boldsymbol{x}^*)^\top \boldsymbol{\lambda}^* = \boldsymbol{0} \quad \Longleftrightarrow \quad Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}^\top.$$

$\blacksquare$

Summary of the proof in one line

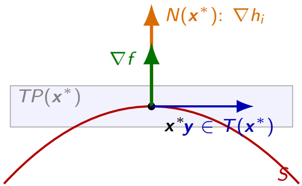
$$\boldsymbol{y} \in T(\boldsymbol{x}^*) \xrightarrow{\text{Thm 20.1 (regularity)}} \text{curve} \xrightarrow{\varphi = f\circ \boldsymbol{x}} \varphi'(t^*) = 0 \Rightarrow \nabla f \perp T(\boldsymbol{x}^*) \xrightarrow{\text{Lemma 20.1}} \nabla f \in N(\boldsymbol{x}^*)$$
$$\Rightarrow \text{multipliers exist.}$$

Reading the theorem geometrically

- At an extremizer, $\nabla f(\boldsymbol{x}^*)$ has **no component inside the tangent space**: moving along the surface cannot decrease f to first order.
- Equivalently, $\nabla f(\boldsymbol{x}^*)$ lies entirely in the normal space — it is "absorbed" by the constraint gradients.
- Compact form of the condition:

$$\nabla f(\boldsymbol{x}^*) \in N(\boldsymbol{x}^*).$$

  If this fails, $\boldsymbol{x}^*$ *cannot* be an extremizer.
- Level set picture: the level set of f through $\boldsymbol{x}^*$ is **tangent** to the surface S at $\boldsymbol{x}^*$.



$\nabla f(\boldsymbol{x}^*)$ is a multiple (combination) of the constraint gradients; it is orthogonal to every tangent direction $\boldsymbol{y}$.

Two warnings to remember

1. The condition is *necessary*, not *sufficient*

There are points satisfying the Lagrange condition that are minimizers, maximizers, or *neither*. Solving the Lagrange equations produces a list of **candidates**; classification needs Theorem 20.4 / 20.5.

2. Regularity cannot be dropped

Example 20.5. Minimize $f(x) = x$ subject to $h(x) = 0$, where

$$h(x) = \begin{cases} x^2, & x < 0 \\ 0, & 0 \le x \le 1 \\ (x-1)^2, & x > 1. \end{cases}$$

The feasible set is $[0, 1]$, so $x^* = 0$ is a local minimizer. But $f'(x^*) = 1$ and $h'(x^*) = 0$, so *no* λ^* satisfies $f' + \lambda^* h' = 0$.

No contradiction: $\nabla h(x^*) = 0$ is *not* linearly independent, so x^* is **not a regular point** and Theorem 20.3 simply does not apply.

Packaging the condition: the Lagrangian

Definition (Lagrangian function)

$\ell : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$,

$$\ell(\boldsymbol{x}, \boldsymbol{\lambda}) \triangleq f(\boldsymbol{x}) + \boldsymbol{\lambda}^\top \boldsymbol{h}(\boldsymbol{x}) = f(\boldsymbol{x}) + \lambda_1 h_1(\boldsymbol{x}) + \cdots + \lambda_m h_m(\boldsymbol{x}).$$

Differentiate with respect to the *whole* argument $[\boldsymbol{x}^\top, \boldsymbol{\lambda}^\top]^\top$:

$$D\ell(\boldsymbol{x}, \boldsymbol{\lambda}) = \left[D_{\boldsymbol{x}}\ell(\boldsymbol{x}, \boldsymbol{\lambda}),\ D_{\boldsymbol{\lambda}}\ell(\boldsymbol{x}, \boldsymbol{\lambda})\right], \qquad \begin{aligned} D_{\boldsymbol{x}}\ell(\boldsymbol{x}, \boldsymbol{\lambda}) &= Df(\boldsymbol{x}) + \boldsymbol{\lambda}^\top D\boldsymbol{h}(\boldsymbol{x}), \\ D_{\boldsymbol{\lambda}}\ell(\boldsymbol{x}, \boldsymbol{\lambda}) &= \boldsymbol{h}(\boldsymbol{x})^\top. \end{aligned}$$

Lagrange condition, compactly

$$D_{\boldsymbol{x}}\ell(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{0}^\top, \quad D_{\boldsymbol{\lambda}}\ell(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{0}^\top \qquad \Longleftrightarrow \qquad \boxed{D\ell(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{0}^\top}$$

The constrained FONC for f is the unconstrained FONC for ℓ. Note that $D_{\boldsymbol{\lambda}}\ell = \boldsymbol{0}^\top$ just reproduces feasibility $\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}$.

Worked example: the ellipse (Example 20.7)

Problem

Extremize $f(\boldsymbol{x}) = x_1^2 + x_2^2$ on the ellipse $\{[x_1, x_2]^\top : \; h(\boldsymbol{x}) = x_1^2 + 2x_2^2 - 1 = 0\}$.

$\nabla f = [2x_1, 2x_2]^\top$, $\nabla h = [2x_1, 4x_2]^\top$. Since $\nabla h = \mathbf{0}$ only at the origin, which is infeasible, **every feasible point is regular**.

$$\ell(\boldsymbol{x}, \lambda) = x_1^2 + x_2^2 + \lambda\,(x_1^2 + 2x_2^2 - 1)$$

$$D_{\boldsymbol{x}}\ell = \mathbf{0}^\top,\; D_\lambda \ell = 0 \implies \begin{cases} 2x_1 + 2\lambda x_1 = 0 & \text{(i)} \\ 2x_2 + 4\lambda x_2 = 0 & \text{(ii)} \\ x_1^2 + 2x_2^2 = 1 & \text{(iii)} \end{cases}$$

Solving. From (i): $x_1 = 0$ or $\lambda = -1$.

- $x_1 = 0$: (iii) gives $x_2 = \pm 1/\sqrt{2}$; then (ii) forces $\lambda = -1/2$.
- $\lambda = -1$: (ii) gives $x_2 = 0$; then (iii) gives $x_1 = \pm 1$.

$$\boldsymbol{x}^{(1)} = \begin{bmatrix} 0 \\ 1/\sqrt{2} \end{bmatrix},\; \boldsymbol{x}^{(2)} = \begin{bmatrix} 0 \\ -1/\sqrt{2} \end{bmatrix},\; \boldsymbol{x}^{(3)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},\; \boldsymbol{x}^{(4)} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}; \quad f(\boldsymbol{x}^{(1,2)}) = \tfrac{1}{2},\; f(\boldsymbol{x}^{(3,4)}) = 1.$$

Four candidates. Which are minimizers? Theorem 20.4 will decide.

Preparing the second-order object

Assume now $f, \boldsymbol{h} \in \mathcal{C}^2$. Let $\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{\lambda})$ be the **Hessian of the Lagrangian with respect to $\boldsymbol{x}$**:

$$\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = D^2_{\boldsymbol{x}} \ell(\boldsymbol{x}, \boldsymbol{\lambda}) = \boldsymbol{F}(\boldsymbol{x}) + \lambda_1 \boldsymbol{H}_1(\boldsymbol{x}) + \cdots + \lambda_m \boldsymbol{H}_m(\boldsymbol{x}),$$

where $\boldsymbol{F}(\boldsymbol{x}) = D^2 f(\boldsymbol{x})$ and

$$\boldsymbol{H}_k(\boldsymbol{x}) = D^2 h_k(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial^2 h_k}{\partial x_1^2}(\boldsymbol{x}) & \cdots & \frac{\partial^2 h_k}{\partial x_n \partial x_1}(\boldsymbol{x}) \\ \vdots & & \vdots \\ \frac{\partial^2 h_k}{\partial x_1 \partial x_n}(\boldsymbol{x}) & \cdots & \frac{\partial^2 h_k}{\partial x_n^2}(\boldsymbol{x}) \end{bmatrix}.$$

Notation

$$[\boldsymbol{\lambda}\boldsymbol{H}(\boldsymbol{x})] \triangleq \lambda_1 \boldsymbol{H}_1(\boldsymbol{x}) + \cdots + \lambda_m \boldsymbol{H}_m(\boldsymbol{x}), \qquad \text{so} \qquad \boxed{\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = \boldsymbol{F}(\boldsymbol{x}) + [\boldsymbol{\lambda}\boldsymbol{H}(\boldsymbol{x})]}$$

$\boldsymbol{L}$ is symmetric, $n \times n$, and plays the role that $\boldsymbol{F}$ played in the *unconstrained* theory.

Theorem 20.4 (Second-Order Necessary Conditions)

Statement

Let $\boldsymbol{x}^*$ be a **local minimizer** of $f : \mathbb{R}^n \to \mathbb{R}$ subject to $\boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}$, $\boldsymbol{h} : \mathbb{R}^n \to \mathbb{R}^m$, $m \leq n$, and $f, \boldsymbol{h} \in \mathcal{C}^2$. Suppose $\boldsymbol{x}^*$ is regular. Then there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ such that:

1. $Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}^\top$;
2. for all $\boldsymbol{y} \in T(\boldsymbol{x}^*)$, $\boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\, \boldsymbol{y} \;\geq\; 0$.

Read part 2 very carefully

$\boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ is required to be positive *semidefinite* **only on the tangent space** $T(\boldsymbol{x}^*)$, *not* on all of $\mathbb{R}^n$.

$$\boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{y} \geq 0 \;\; \forall \boldsymbol{y} \in T(\boldsymbol{x}^*) \quad \text{is much weaker than} \quad \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) \succeq 0.$$

Directions that leave the surface are irrelevant: we are never allowed to move in them.

Proof of Theorem 20.4 — plan

Part 1 is exactly Theorem 20.3 (a local minimizer at a regular point), so $\boldsymbol{\lambda}^*$ exists with $Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \mathbf{0}^\top$. **Only part 2 remains.**

Plan

Fix $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ and produce a *twice*-differentiable curve on S realising $\boldsymbol{y}$. Then extract *two* pieces of information at t^*:

(A) from $\varphi(t) = f(\boldsymbol{x}(t))$ having a local min: $\ddot{\varphi}(t^*) \geq 0$;

(B) from the identity $\boldsymbol{\lambda}^{*\top}\boldsymbol{h}(\boldsymbol{x}(t)) \equiv 0$: a second derivative equal to 0.

Each of (A), (B) contains an unwanted term involving the *acceleration* $\ddot{\mathbf{x}}(t^*)$ — a quantity we do not control. **Adding (A) and (B) makes those two terms combine into $\big(Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*)\big)\ddot{\mathbf{x}}(t^*)$, which is 0 by part 1.**

That cancellation is the entire trick of the proof — and it is why the multiplier $\boldsymbol{\lambda}^*$ from part 1 is exactly the right thing to insert in (B).

Proof of Theorem 20.4 — Step 1: the curve

Step 1

Let $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ be arbitrary. Since $\boldsymbol{x}^*$ is regular and $\boldsymbol{h} \in \mathcal{C}^2$, the argument of Theorem 20.1 (Implicit Function Theorem) yields a **twice-differentiable** curve $\{\boldsymbol{x}(t) : t \in (a, b)\}$ on S with

$$\boldsymbol{h}(\boldsymbol{x}(t)) = \mathbf{0} \ \ \forall t, \qquad \boldsymbol{x}(t^*) = \boldsymbol{x}^*, \qquad \dot{\boldsymbol{x}}(t^*) = \boldsymbol{y}.$$

Step 2

Set $\varphi(t) = f(\boldsymbol{x}(t))$. As in Theorem 20.3, t^* is an unconstrained local minimizer of φ. By the one-variable *second-order* necessary condition (Theorem 6.2), $\dfrac{d^2\varphi}{dt^2}(t^*) \ \geq \ 0.$

A product rule we shall use twice

For differentiable $\boldsymbol{y}(\cdot), \boldsymbol{z}(\cdot) : \mathbb{R} \to \mathbb{R}^n$, $\dfrac{d}{dt}\big(\boldsymbol{y}(t)^\top \boldsymbol{z}(t)\big) = \boldsymbol{z}(t)^\top \dfrac{d\boldsymbol{y}}{dt}(t) + \boldsymbol{y}(t)^\top \dfrac{d\boldsymbol{z}}{dt}(t).$

Proof of Theorem 20.4 — Step 3: computing $\ddot{\varphi}(t^*)$

First derivative (chain rule): $\dfrac{d\varphi}{dt}(t) = Df(\boldsymbol{x}(t))\,\dot{\boldsymbol{x}}(t) = \nabla f(\boldsymbol{x}(t))^\top \dot{\boldsymbol{x}}(t)$.

Differentiate again with the product rule, using $\dfrac{d}{dt}\nabla f(\boldsymbol{x}(t)) = \boldsymbol{F}(\boldsymbol{x}(t))\dot{\boldsymbol{x}}(t)$:

$$\frac{d^2\varphi}{dt^2}(t) = \underbrace{\dot{\boldsymbol{x}}(t)^\top \boldsymbol{F}(\boldsymbol{x}(t))\,\dot{\boldsymbol{x}}(t)}_{\text{curvature of } f \text{ along the curve}} + \underbrace{Df(\boldsymbol{x}(t))\,\ddot{\boldsymbol{x}}(t)}_{\text{effect of the curve bending}} .$$

At $t = t^*$, with $\boldsymbol{x}(t^*) = \boldsymbol{x}^*$, $\dot{\boldsymbol{x}}(t^*) = \boldsymbol{y}$:

Inequality (A)

$$\boldsymbol{y}^\top \boldsymbol{F}(\boldsymbol{x}^*)\,\boldsymbol{y} + Df(\boldsymbol{x}^*)\,\ddot{\boldsymbol{x}}(t^*) \;\geq\; 0.$$

Why we cannot stop here

The term $Df(\boldsymbol{x}^*)\ddot{\boldsymbol{x}}(t^*)$ depends on *which* curve we chose, not just on $\boldsymbol{y}$. It must be eliminated. That is the job of Step 4.

Proof of Theorem 20.4 — Step 4: differentiating the constraint twice

Since $\boldsymbol{h}(\boldsymbol{x}(t)) = \boldsymbol{0}$ for all t, the scalar function $t \mapsto \boldsymbol{\lambda}^{*\top}\boldsymbol{h}(\boldsymbol{x}(t))$ is *identically zero*; hence all of its derivatives vanish: $\dfrac{d^2}{dt^2}\left[\boldsymbol{\lambda}^{*\top}\boldsymbol{h}(\boldsymbol{x}(t))\right] = 0$. Compute it carefully:

$$\begin{aligned}
\frac{d^2}{dt^2}\boldsymbol{\lambda}^{*\top}\boldsymbol{h}(\boldsymbol{x}(t)) &= \frac{d}{dt}\left[\boldsymbol{\lambda}^{*\top}\frac{d}{dt}\boldsymbol{h}(\boldsymbol{x}(t))\right] = \frac{d}{dt}\left[\sum_{k=1}^{m}\lambda_k^*\frac{d}{dt}h_k(\boldsymbol{x}(t))\right] \\
&= \frac{d}{dt}\left[\sum_{k=1}^{m}\lambda_k^*\, Dh_k(\boldsymbol{x}(t))\,\dot{\boldsymbol{x}}(t)\right] = \sum_{k=1}^{m}\lambda_k^*\frac{d}{dt}\left(Dh_k(\boldsymbol{x}(t))\dot{\boldsymbol{x}}(t)\right) \\
&= \sum_{k=1}^{m}\lambda_k^*\left[\dot{\boldsymbol{x}}(t)^\top \boldsymbol{H}_k(\boldsymbol{x}(t))\dot{\boldsymbol{x}}(t) + Dh_k(\boldsymbol{x}(t))\ddot{\boldsymbol{x}}(t)\right] \\
&= \dot{\boldsymbol{x}}(t)^\top\left[\boldsymbol{\lambda}^*\boldsymbol{H}(\boldsymbol{x}(t))\right]\dot{\boldsymbol{x}}(t) + \boldsymbol{\lambda}^{*\top}D\boldsymbol{h}(\boldsymbol{x}(t))\,\ddot{\boldsymbol{x}}(t) \;=\; 0.
\end{aligned}$$

Equation (B) — evaluate at $t = t^*$

$$\boldsymbol{y}^\top\left[\boldsymbol{\lambda}^*\boldsymbol{H}(\boldsymbol{x}^*)\right]\boldsymbol{y} + \boldsymbol{\lambda}^{*\top}D\boldsymbol{h}(\boldsymbol{x}^*)\,\ddot{\boldsymbol{x}}(t^*) \;=\; 0.$$

Proof of Theorem 20.4 — Step 5: add (A) and (B)

$$\begin{array}{lrcl} \text{(A)} & \boldsymbol{y}^\top \boldsymbol{F}(\boldsymbol{x}^*)\boldsymbol{y} + Df(\boldsymbol{x}^*)\ddot{\boldsymbol{x}}(t^*) & \geq & 0 \\ \text{(B)} & \boldsymbol{y}^\top [\boldsymbol{\lambda}^* \boldsymbol{H}(\boldsymbol{x}^*)]\boldsymbol{y} + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*)\ddot{\boldsymbol{x}}(t^*) & = & 0 \end{array}$$

$$\text{(A)+(B)} \quad \boldsymbol{y}^\top \Big(\boldsymbol{F}(\boldsymbol{x}^*) + [\boldsymbol{\lambda}^* \boldsymbol{H}(\boldsymbol{x}^*)]\Big)\boldsymbol{y} + \Big(Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*)\Big)\ddot{\boldsymbol{x}}(t^*) \ \geq\ 0.$$

By **part 1** (Lagrange's theorem) the bracket multiplying $\ddot{\boldsymbol{x}}(t^*)$ vanishes:

$$Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}^\top \quad \Longrightarrow \quad \big(Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*)\big)\ddot{\boldsymbol{x}}(t^*) = 0.$$

Therefore, recalling $\boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{F}(\boldsymbol{x}^*) + [\boldsymbol{\lambda}^* \boldsymbol{H}(\boldsymbol{x}^*)]$,

$$\boxed{\boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\,\boldsymbol{y} \ \geq\ 0}$$

and since $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ was arbitrary, part 2 holds. ∎

Post-mortem: what each hypothesis did

- $\boldsymbol{x}^*$ **regular** $\Rightarrow$ Theorem 20.1 gives, for *every* $\boldsymbol{y} \in T(\boldsymbol{x}^*)$, a curve on S with velocity $\boldsymbol{y}$. Without it, $T(\boldsymbol{x}^*)$ could be strictly larger than the set of genuine tangent directions, and the conclusion may fail (Example 20.5).
- $\boldsymbol{h} \in \mathcal{C}^2 \Rightarrow$ the curve can be taken *twice* differentiable, so $\ddot{\boldsymbol{x}}$ exists; $f \in \mathcal{C}^2 \Rightarrow \boldsymbol{F}$ exists and $\ddot{\varphi}$ can be computed.
- $\boldsymbol{x}^*$ **a local minimizer** $\Rightarrow$ the inequality $\ddot{\varphi}(t^*) \geq 0$ (for a maximizer the inequality reverses: $\boldsymbol{y}^\top \boldsymbol{L} \boldsymbol{y} \leq 0$ on $T(\boldsymbol{x}^*)$).
- **The multiplier $\boldsymbol{\lambda}^*$ of part 1** is what makes the acceleration terms cancel. The theorem would be false with an arbitrary $\boldsymbol{\lambda}$.

Analogy with the unconstrained case

$$\text{unconstrained: } \nabla f(\boldsymbol{x}^*) = \boldsymbol{0}, \quad \boldsymbol{y}^\top \boldsymbol{F}(\boldsymbol{x}^*)\boldsymbol{y} \geq 0 \ \ \forall \boldsymbol{y} \in \mathbb{R}^n$$

$$\text{constrained: } \nabla_{\boldsymbol{x}} \ell = \boldsymbol{0}, \quad \boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{y} \geq 0 \ \ \forall \boldsymbol{y} \in T(\boldsymbol{x}^*)$$

$$(f \rightsquigarrow \ell, \quad \mathbb{R}^n \rightsquigarrow T(\boldsymbol{x}^*).)$$

The companion result: Theorem 20.5 (sufficient conditions)

Theorem 20.5 (Second-Order Sufficient Conditions)

Suppose $f, \boldsymbol{h} \in \mathcal{C}^2$ and there exist $\boldsymbol{x}^* \in \mathbb{R}^n$, $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ such that

1. $Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}^\top$ (and $\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}$);
2. $\boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{y} > 0$ for all $\boldsymbol{y} \in T(\boldsymbol{x}^*)$, $\boldsymbol{y} \neq \boldsymbol{0}$.

Then $\boldsymbol{x}^*$ is a **strict local minimizer** of f subject to $\boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{0}$.

- Note the differences from Theorem 20.4: the inequality is *strict*, and **regularity is not assumed** — here $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ is *given*, not deduced.
- Logical direction: 20.4 goes "minimizer $\Rightarrow$ conditions"; 20.5 goes "conditions $\Rightarrow$ minimizer".
- The gap between ≥ 0 and > 0 is genuine — as in one-variable calculus, where $f''(x^*) = 0$ decides nothing.

Example: classifying the four candidates on the ellipse

Recall $f(\boldsymbol{x}) = x_1^2 + x_2^2$, $h(\boldsymbol{x}) = x_1^2 + 2x_2^2 - 1$. Then

$$\boldsymbol{F}(\boldsymbol{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \boldsymbol{H}(\boldsymbol{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \qquad \boldsymbol{L}(\boldsymbol{x}, \lambda) = \begin{bmatrix} 2+2\lambda & 0 \\ 0 & 2+4\lambda \end{bmatrix}.$$

Candidate $\boldsymbol{x}^{(1)} = [0, 1/\sqrt{2}]^\top$, $\lambda^* = -\frac{1}{2}$.

$$\nabla h(\boldsymbol{x}^{(1)}) = [0,\ 2\sqrt{2}]^\top \ \Rightarrow\ T(\boldsymbol{x}^{(1)}) = \{\boldsymbol{y} : 2\sqrt{2}\, y_2 = 0\} = \{[\alpha, 0]^\top : \alpha \in \mathbb{R}\},$$

$$\boldsymbol{L} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \boldsymbol{y}^\top \boldsymbol{L} \boldsymbol{y} = \alpha^2 \ > \ 0 \ \text{ for } \boldsymbol{y} \neq \boldsymbol{0}.$$

Theorem 20.4 is satisfied; in fact Theorem 20.5 applies: $\boldsymbol{x}^{(1)}$ is a strict local minimizer.

(Identically for $\boldsymbol{x}^{(2)} = [0, -1/\sqrt{2}]^\top$.)

Example (continued): rejecting the other two candidates

Candidate $\boldsymbol{x}^{(3)} = [1, 0]^\top$, $\lambda^* = -1$.

$$\nabla h(\boldsymbol{x}^{(3)}) = [2, 0]^\top \;\Rightarrow\; T(\boldsymbol{x}^{(3)}) = \{\boldsymbol{y} : 2y_1 = 0\} = \{[0, \beta]^\top : \beta \in \mathbb{R}\},$$

$$\boldsymbol{L}(\boldsymbol{x}^{(3)}, -1) = \begin{bmatrix} 2-2 & 0 \\ 0 & 2-4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}, \qquad \boldsymbol{y}^\top \boldsymbol{L} \boldsymbol{y} = -2\beta^2 \;<\; 0 \;\; (\beta \neq 0).$$

Conclusion

Part 2 of Theorem 20.4 *fails*, so $\boldsymbol{x}^{(3)}$ is **not** a local minimizer (likewise $\boldsymbol{x}^{(4)} = [-1, 0]^\top$). The reversed inequality $\boldsymbol{y}^\top \boldsymbol{L} \boldsymbol{y} \leq 0$ on T holds, consistent with $\boldsymbol{x}^{(3)}, \boldsymbol{x}^{(4)}$ being maximizers.

Values: $f(\boldsymbol{x}^{(1)}) = f(\boldsymbol{x}^{(2)}) = \frac{1}{2}$ (minimum), $f(\boldsymbol{x}^{(3)}) = f(\boldsymbol{x}^{(4)}) = 1$ (maximum).

Geometric check: the ellipse $x_1^2 + 2x_2^2 = 1$ has semi-axes 1 (along x_1) and $1/\sqrt{2}$ (along x_2); f is squared distance from the origin. ✓

Example: the box of maximum volume (Example 20.6)

Problem

With a fixed area A of cardboard, build a closed box of maximum volume:

$$\text{maximize } x_1x_2x_3 \quad \text{subject to } x_1x_2 + x_2x_3 + x_3x_1 = \frac{A}{2}.$$

Write $f(\boldsymbol{x}) = -x_1x_2x_3$, $h(\boldsymbol{x}) = x_1x_2 + x_2x_3 + x_3x_1 - A/2$. The Lagrange condition gives

$$x_2x_3 - \lambda(x_2 + x_3) = 0, \quad x_1x_3 - \lambda(x_1 + x_3) = 0, \quad x_1x_2 - \lambda(x_1 + x_2) = 0,$$

together with the constraint.

- If some $x_i = 0$ the equations force $x_jx_k = 0$, contradicting the constraint; so all $x_i \neq 0$. Similarly $\lambda \neq 0$ (adding the three equations with $\lambda = 0$ gives $x_1x_2 + x_2x_3 + x_3x_1 = 0$).
- Multiply the first equation by x_1, the second by x_2 and subtract: $x_3\lambda(x_1 - x_2) = 0 \Rightarrow x_1 = x_2$; similarly $x_2 = x_3$.
- Constraint $\Rightarrow x_1 = x_2 = x_3 = \sqrt{A/6}$: **the cube**.

Example: a generalized eigenvalue problem (Example 20.8)

Problem

Maximize the Rayleigh quotient $\dfrac{\boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x}}{\boldsymbol{x}^\top \boldsymbol{P}\boldsymbol{x}}$ with $\boldsymbol{Q} = \boldsymbol{Q}^\top \succeq 0$, $\boldsymbol{P} = \boldsymbol{P}^\top \succ 0$.

The quotient is invariant under $\boldsymbol{x} \mapsto t\boldsymbol{x}$ $(t \neq 0)$, so normalise: $\boldsymbol{x}^\top \boldsymbol{P}\boldsymbol{x} = 1$. With $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x}$ and $h(\boldsymbol{x}) = 1 - \boldsymbol{x}^\top \boldsymbol{P}\boldsymbol{x}$,

$$\ell(\boldsymbol{x}, \lambda) = \boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x} + \lambda(1 - \boldsymbol{x}^\top \boldsymbol{P}\boldsymbol{x}), \qquad D_{\boldsymbol{x}}\ell = 2\boldsymbol{x}^\top \boldsymbol{Q} - 2\lambda \boldsymbol{x}^\top \boldsymbol{P} = \boldsymbol{0}^\top.$$

Hence $\boldsymbol{Q}\boldsymbol{x} = \lambda \boldsymbol{P}\boldsymbol{x}$, i.e. $\boldsymbol{P}^{-1}\boldsymbol{Q}\boldsymbol{x} = \lambda \boldsymbol{x}$:

- $\boldsymbol{x}^*$ is an **eigenvector** of $\boldsymbol{P}^{-1}\boldsymbol{Q}$ and λ^* the corresponding eigenvalue;
- from $\boldsymbol{x}^{*\top} \boldsymbol{P}\boldsymbol{x}^* = 1$: $\lambda^* = \boldsymbol{x}^{*\top} \boldsymbol{Q}\boldsymbol{x}^* = f(\boldsymbol{x}^*)$, so the maximum value *is* the largest (generalized) eigenvalue.

A classical linear-algebra fact falls out of Lagrange's theorem.

A recipe you can follow in the exam

1. **Set up.** Write the problem as $\min f$ s.t. $\boldsymbol{h} = \mathbf{0}$ (negate for maximization).
2. **Check regularity.** Compute $D\boldsymbol{h}(\boldsymbol{x})$ and identify feasible points where $\operatorname{rank} D\boldsymbol{h} < m$ — these must be examined *separately*, since Theorem 20.3 says nothing there.
3. **Form** $\ell(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\top \boldsymbol{h}(\boldsymbol{x})$.
4. **Solve** $D_{\boldsymbol{x}}\ell = \mathbf{0}^\top$, $\boldsymbol{h}(\boldsymbol{x}) = \mathbf{0}$: $n + m$ equations, $n + m$ unknowns. List all solutions $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ — these are the *candidates*.
5. **Compute the tangent space** at each candidate: $T(\boldsymbol{x}^*) = \mathcal{N}(D\boldsymbol{h}(\boldsymbol{x}^*))$; get a basis.
6. **Compute** $\boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \boldsymbol{F}(\boldsymbol{x}^*) + \sum_k \lambda_k^* \boldsymbol{H}_k(\boldsymbol{x}^*)$ and test the sign of $\boldsymbol{y}^\top \boldsymbol{L} \boldsymbol{y}$ on $T(\boldsymbol{x}^*)$ only (substitute the basis).
7. **Conclude.** > 0 on $T(\boldsymbol{x}^*) \setminus \{\mathbf{0}\}$: strict local min (Thm 20.5). Some $\boldsymbol{y} \in T(\boldsymbol{x}^*)$ with < 0: not a local min (Thm 20.4). ≥ 0 with equality somewhere: inconclusive — investigate directly.

Common mistakes

Six errors to avoid

1. Forgetting to *verify regularity* before applying Theorem 20.3.
2. Treating the Lagrange condition as sufficient — it only produces candidates.
3. Testing $\boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) \succeq 0$ on all of $\mathbb{R}^n$ instead of on $T(\boldsymbol{x}^*)$. (This is a *stronger* requirement and will reject genuine minimizers.)
4. Using the wrong $\boldsymbol{\lambda}$ in $\boldsymbol{L}$ — it must be the multiplier attached to *that* candidate $\boldsymbol{x}^*$.
5. Forgetting the constraint equation $\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}$ when counting equations.
6. Sign conventions: with $\ell = f + \boldsymbol{\lambda}^\top \boldsymbol{h}$ the condition is $\nabla f + D\boldsymbol{h}^\top \boldsymbol{\lambda} = \boldsymbol{0}$; using $\ell = f - \boldsymbol{\lambda}^\top \boldsymbol{h}$ merely flips the sign of $\boldsymbol{\lambda}$ — be consistent within a problem.

Exercises to attempt

1. Minimize $f(\boldsymbol{x}) = x_1^2 + x_2^2 + x_3^2$ subject to $x_1 + x_2 + x_3 = 3$. Find $\boldsymbol{x}^*, \lambda^*$, compute $T(\boldsymbol{x}^*)$, and verify Theorem 20.4.
2. For $f(\boldsymbol{x}) = x_1 x_2$ on the circle $x_1^2 + x_2^2 = 1$: find all four candidates and classify each using $\boldsymbol{L}$ restricted to the tangent line.
3. Let $S = \{\boldsymbol{x} \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1,\ x_1 + x_2 + x_3 = 0\}$ $(m = 2)$. Show every point of S is regular, compute $T(\boldsymbol{x}^*)$ at $\boldsymbol{x}^* = \frac{1}{\sqrt{2}}[1, -1, 0]^\top$, and extremize $f(\boldsymbol{x}) = x_3$ on S.
4. Give an example of a feasible point where $\boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ is *indefinite* on $\mathbb{R}^n$ but positive definite on $T(\boldsymbol{x}^*)$, and $\boldsymbol{x}^*$ is a strict local minimizer.
5. Prove that if $\boldsymbol{x}^*$ is a local *maximizer* (regular, $f, \boldsymbol{h} \in \mathcal{C}^2$) then $\boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{y} \leq 0$ for all $\boldsymbol{y} \in T(\boldsymbol{x}^*)$. *(Hint: apply Theorem 20.4 to $-f$.)*

Summary

The chain of ideas

$$\text{regular point} \;\Rightarrow\; T(\boldsymbol{x}^*) = \mathcal{N}(D\boldsymbol{h}(\boldsymbol{x}^*)),\; N(\boldsymbol{x}^*) = \mathcal{R}(D\boldsymbol{h}(\boldsymbol{x}^*)^\top), \quad T(\boldsymbol{x}^*)^\perp = N(\boldsymbol{x}^*)$$

$$\text{Thm 20.1: tangent vectors} = \text{velocities of curves on } S$$

$$\text{Thm 20.3: } \varphi'(t^*) = 0\ \forall\text{curves} \;\Rightarrow\; \nabla f(\boldsymbol{x}^*) \in N(\boldsymbol{x}^*) \;\Rightarrow\; Df(\boldsymbol{x}^*) + \boldsymbol{\lambda}^{*\top} D\boldsymbol{h}(\boldsymbol{x}^*) = \boldsymbol{0}^\top$$

$$\text{Thm 20.4: } \varphi''(t^*) \geq 0 \;+\; \tfrac{d^2}{dt^2}\left[\boldsymbol{\lambda}^{*\top}\boldsymbol{h}(\boldsymbol{x}(t))\right] = 0 \;\Rightarrow\; \boldsymbol{y}^\top \boldsymbol{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{y} \geq 0 \;\; \forall \boldsymbol{y} \in T(\boldsymbol{x}^*)$$

- Both theorems are **necessary** conditions at a **regular** point.
- Both are proved by the same device: *reduce to one variable along a curve inside the feasible set, then use ordinary calculus.*
- The Lagrangian $\ell = f + \boldsymbol{\lambda}^\top \boldsymbol{h}$ makes the constrained conditions look exactly like the unconstrained ones, with $\mathbb{R}^n$ replaced by $T(\boldsymbol{x}^*)$.

Thank You!

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